

CHRISTIAAN HUYGENS'

The Pendulum Clock

or

*Geometrical Demonstrations
Concerning the Motion
of Pendula
as Applied to Clocks*



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INTRODUCTION BY

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A M E S

tory motions of the pendulum cannot disturb the instrument, and nothing would destroy the motion of the pendulum more than this. Next the width of the axes CC and FG, which are one inch, and the lead weight suspended below, completely remove all freedom of movement from the clock and have the effect of always returning it to rest and to a perpendicular orientation if it had been disturbed by some strong and sudden shifting of the ship.

Instruments which have been modified in this way have yet to be taken to sea and put to the test. But they offer an almost certain hope of success because, in experiments conducted so far, they have been found to hold up under every kind of motion much better than prior clocks.

PART II

The Falling of Heavy Bodies and Their Motion in a Cycloid

HYPOTHESES

I

If there were no gravity, and if the air did not impede the motion of bodies, then any body will continue its given motion with uniform velocity in a straight line.

II

By the action of gravity, whatever its sources,¹ it happens that bodies are moved by a motion composed both of a uniform motion in one direction or another and of a motion downward due to gravity.

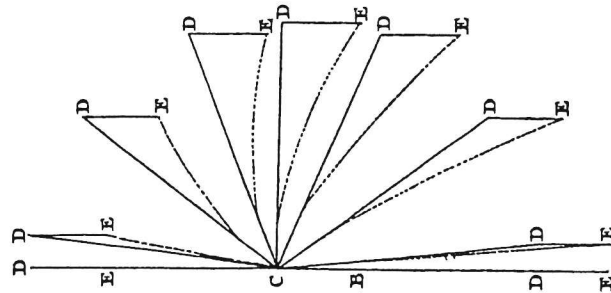
III

These two motions can be considered separately, with neither being impeded by the other.



LET C be a heavy body which, starting from rest, crosses the distance CB by the force of gravity in a certain time F [Fig. 7]. And let the same body be imagined to undergo another motion by which, assuming that gravity does not exist, it crosses the straight line CD with a uniform motion in the same time F. When the force of gravity is added, the body will not move from C to D in the time F

For various earlier writings by Huygens on topics in Part II, see *Oeuvres* 17:125-41.
1. In the *Horologium oscillatorium* Huygens restricts himself to the phenomenological level and avoids any discussion of the cause of gravity. But at that time he was a strong advocate of the Cartesian account of gravity as caused by the vortex motion of subtle matter, a view which he presented and defended against strong opposition at a meeting of the Académie des Sciences in 1668, and which is found in his *Discours de la cause de la pesanteur* (1668; published in 1690). (*Oeuvres* 21:445-88.)



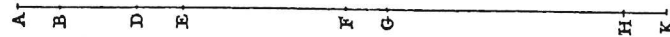
[Fig. 7]

but rather to some point E vertically below D such that the distance DE equals the distance CB. And thus the uniform motion and the motion due to gravity each make a contribution, and neither impedes the other. In what follows later we will define the line in which the body moves with this composite motion when the uniform motion is neither straight up or down but in an oblique direction. But when the uniform motion CD occurs downward on the perpendicular, it is obvious that the line CD is increased by the straight line DE when the motion due to gravity is added. Likewise, when the uniform motion CD is directed upward, CD is decreased by the straight line DE, so that, for example, after the time F the body will always be found at the point E. Thus, if we consider the two motions separately in each case, as we said, and if we recognize that neither motion is in any way impeded by the other, then from this we can discover the cause and the laws of acceleration of heavy falling bodies. And first we will show the following two things.

PROPOSITION I

In equal times equal amounts of velocity are added to a falling body, and in equal times the distances crossed by a body falling from rest are successively increased by an equal amount.

[Fig. 8]



Let there be a heavy body at rest at A [Fig. 8]. In the first unit of time it falls through the distance AB; and when it has arrived at B, it has acquired a velocity by which it next could cross the distance BD with a uniform velocity in the second unit of time. But we know that in the second unit of time it will cross a distance greater than BD because it would travel the distance BD only if all the action of gravity had ceased at B. Actually it moves with a motion composed of the uniform motion by which it would have crossed the distance BD and of a motion characteristic of falling bodies by which it necessarily falls through a distance equal to AB. Hence by adding DE, which is equal to AB, to BD, we know that in the second unit of time the body will arrive at E.

But if we inquire what velocity the body has at E at the end of the second unit of time, we find that this ought to be double the velocity which it had at B at the end of the first unit of time. For we said that it is moved by a motion composed of a uniform motion equal to the velocity acquired at B and of a motion due to gravity, which clearly is the same in the second unit of time as in the first. Hence a velocity ought to be added to the falling body in the second unit of time which is equal to the velocity added in the first unit of time. Thus, since it conserves the whole velocity acquired at the end of the first unit of time, it is clear that at the end of the second unit of time it has twice, or double, the velocity which it acquired at the end of the first unit of time.

Now if, after having arrived at E, the body were to be moved with a uniform velocity equal to what it has acquired at E, it is clear that in a third unit of time equal to each of the first two it would cross the distance EF, which is double the distance BD. For we said that the latter was crossed with half this velocity by a uniform motion in an equal time. But by adding again the action of gravity, in the third unit of time the body will cross the distance EF and also the distance FG, which is equal to AB or to DE. And thus at the end of the third unit of time the body will be found at G. It will have here a velocity which is triple that which it had at B at the end of the first unit of time. For in addition to the velocity acquired at E, which we said was double that acquired at B, in the third unit of time of the fall a velocity is added which again is equal to the velocity at the end of the first unit of time. Hence at the end of the third unit of time both velocities add up to triple the velocity found at the end of the first unit of time.

In the same way it can be shown that in the fourth unit of time the body ought to cross both the distance GH, which is triple BD, and the distance HK, which is equal to AB; and the velocity at K, at the end of the fourth unit of time, will be quadruple what it was at B at the end of the first unit of time. Therefore, it is clear that whatever distances we take to be crossed successively in equal times, these distances will each increase by an amount equal to BD, and simultaneously the velocities will also be increased equally in equal times.

PROPOSITION II

The distance crossed in a certain time by a body beginning to fall from rest is one-half the distance which it would cross in an equal time with a uniform motion whose velocity is equal to the velocity acquired at the last moment of the fall.

Assume what was stated in the prior proposition where AB was the distance crossed in a certain time by a body falling from A [Fig. 8]. BD was the distance which in an equal time the body was understood to have crossed with a uniform velocity equal to the velocity which was acquired at the end of the first unit of time or at the end of the distance AB. Now I say that the distance BD is twice AB.

The distances crossed by a falling body in the first four equal times are AB, BE, EG, and GH. There is a certain proportion between these distances. Let us suppose times double these times so that, for example, for the first time we take the two times in which the distances AB and BE were crossed, and for the second time we take the two remaining times in which the distances EG and GK were crossed. It is then necessary that the distances AE and EK, which are crossed from rest in equal times, are related to each other as are the distances AB and BE, which are also crossed from rest in equal times.

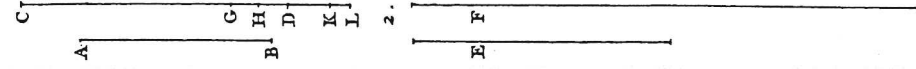
From this it follows that AB is to BE as AE is to EK. By conversion KE is to EA as EB or DA is to AB. Also by division DB is to BA as the excess of KE over AE is to EA. From what was proven in the prior proposition, it follows that KE is equal to twice AB plus five times BD. But EA is equal to twice AB plus BD. Thus it is clear that the mentioned excess of KE over EA is equal to four times BD. Hence DB will be to BA as four times DB is to EA. Thus EA will be four times BA. But EA is equal, as we said, to twice AB plus BD. Therefore BD is equal to twice AB. Q.E.D.

PROPOSITION III

If two distances are crossed by a falling body in any times, each of which is measured from the beginning of the fall, these distances are related to each other as the duplicate ratio of these times, or as the squares of the times, or as the squares of the velocities acquired at the end of these times.

For it was shown in the preceding proposition that the distances AB, BE, EG, and GK [Fig. 8], whatever their size, which are crossed in equal times by a falling body, increase by the same amount, which is equal to BD. Now it is clear that BD is twice AB, that BE is triple AB, that EG is five times AB, that GK is seven times AB, and that the remaining distances are

[Fig. 9]



successively increased according to the progression of odd numbers beginning from one, i.e., 1, 3, 5, 7, 9, etc. And since any sum of these numbers taken consecutively, makes a square whose side equals the number of numbers taken (for example, if the first three are added, they make nine; if four sixteen), it follows from this that the distances crossed by a falling body each of which is taken from the beginning of the fall, are related to each other as the duplicate ratio of the times during which the fall occurs, indeed the times are assumed to be commensurable.

But it is easy to extend this demonstration to incommensurable times. Let there be two such times whose ratio is expressed by the lines AB and CD [Fig. 9], and let the distances crossed in these times be E and F, each taken from the beginning of the fall. Now I say that the distance E is related to F as the square of AB is related to the square of CD.

For if this be denied, the first possibility is that the ratio of E to F is greater than the ratio of the square of AB to the square of CD, that is, it has the ratio of the square of AB to the square of CG, where CG is less than CD. From CD let there be subtracted the part DH, which is smaller than DG, the excess of CD over CG, and let this be done in such a way that the remainder HC is commensurable to AB. It is clear that it is possible to do this. Hence CH will be larger than CG. And the square of the time AB will be related to the square of the time CH as the distance E, which is crossed in the time AB, is related to the distance crossed in the time CH, as shown above. But the distance F which is crossed in the time CD is larger than the distance crossed in the time CH. Thus the ratio of the distance E to the distance F is less than the ratio of the square of AB to the square of CH. But it was shown that E is to F as the square of AB is to the square of CG. Therefore, the ratio of the square of AB to the square of CG will be less than the ratio of the square of CG to the square of CH. From this it follows that the square of CG is greater than the square of CH, which is absurd because CH was said to be greater than CG. Therefore it is not the case that the ratio of the distance E to the distance F is greater than the ratio of the square of AB to the square of CD.

The second possibility is that the ratio of E to F is less than the ratio of the square of AB to the square of CD. Let the ratio of E to F be the same as the ratio of the square of AB to the square of CL, where CL is greater than CD. From CL let there be subtracted the line LK which is less than the excess LD by which CD is surpassed by CL, and let this be done in such a way that the remainder KC is commensurable to AB. Hence the square of the time AB is related to the square of the time CK as the distance E, which is crossed in the time AB, is related to the distance crossed in the time CK. But the distance F which is crossed in the time CD is less than this distance. Hence the ratio of E to F is greater than the ratio of the square of AB to the square of CK. But it was shown that E is to F as the square of AB is to the

square of CL. Therefore the ratio of the square of AB to the square of CL is the same as the ratio of the square of AB to the square of CK. Thus the square of CL is less than the square of CK, which is absurd because SL is larger than CK. Therefore the ratio of E to F is not less than the ratio of the square of AB to the square of CD, but they necessarily are the same. Finally, since the velocities acquired at the end of the times AB and CD are related to each other in the same way as these times, it is obvious that E is related to F by the same ratio as the squares of the times AB and CD in which they are crossed. Thus the proposition is established.

PROPOSITION IV

If a heavy body begins to move upwards with the same velocity acquired at the end of a descent, then in equal parts of time it will cross the same distances upwards as it did downwards, and it will rise to the same height from which it descended. Also in equal parts of time it will lose equal amounts of velocity.

As in Proposition 2 let there be any number of distances crossed in equal parts of time by a body falling from rest. Let AB [Fig. 8] be the first distance; the second will be composed of BD, which was crossed with the uniform velocity acquired through AB, and of DE which is equal to AB. The third will be composed of EF, which is twice BD, and of FG which is equal to AB; the fourth will be composed of GH, which is triple BD, and of HK which is again equal to AB; and the subsequent distances, as many as wished, will be increased by the same ratio. Now I say that in as many equal times the same individual distances KG, GE, EB, and BA are crossed successively by a body which moves upwards beginning with the velocity acquired at K at the end of the descent.

For the sake of brevity each velocity will be successively designated by the length of the distance crossed by a body in uniform motion with that velocity in one part of time, just as we described it in the case of descent.

From what was shown in the above-mentioned proposition, when the heavy body has arrived at K, it has a velocity of GH plus BD, that is, the velocity KF, because KF equals HG plus BD. For each part, HK and FG, is equal to AB, and together they equal BD, which is twice AB, as was shown in Proposition 2. Thus, if the velocity acquired at K at the end of the descent is directed upwards, and if the body is moved with a uniform motion, it will cross the distance KF in one part of time. But, after taking account of the action of gravity, the ascent KF will be decreased by the distance FG which is equal to AB, as is clear from what was said in our initial hypothesis. Hence in the first part of time the body rises only

through KG, which is the same distance it descended in the previous part of time. Simultaneously the velocity must have decreased by the same amount which it acquired by falling downwards in one part of time, namely, BD. Thus when the body has risen to the point G, its remaining velocity is HG since at the beginning of the ascent its velocity was HG plus BD. But HG is equal to GD because HG is equal to FE plus DB, that is, HG equals FE plus twice AB, that is, HG equals FE plus FG plus ED. Hence if the body move upwards from G with the uniform velocity which it has at that point, it will cross the distance GD in one part of time. But, after taking account of the action of gravity, this ascent is decreased by the distance DE, which is equal to AB. Thus in this second part of time it ascends through the distance GE which it crossed in the same part of time when falling. Simultaneously the velocity must have decreased by the same amount which it acquired by falling in one part of time, namely, BD. Hence when it has risen up to the point E it has only the velocity FE, which is the remainder when BD is subtracted from GD. For, as we said, BD equals DE plus FG.

But EA equals FE, because FE equals BD taken twice; that is, FE equals BD plus twice AB; that is, FE equals BD plus AB plus DE. Therefore if the body moves upwards from E with the uniform velocity which it has at that point, it will cross the distance EA in one part of time. But, after taking account of the action of gravity, this ascent will be decreased by the distance AB. Hence in this part of time it rises only through the distance EB, which it also crossed in the same part of time by falling. Again the velocity at this point must have decreased by the amount which it acquired by falling in one part of time, namely, BD. Thus when the body has risen up to the point B, its remaining velocity is BD, since at E it had the velocity FE which is twice BD. Hence if the body moves upwards from B with the uniform velocity which it has at that point, in one part of time it will cross a distance equal to DB, that is, twice AB. But, after taking account of the action of gravity, this ascent is decreased by a distance equal to AB. Hence in this part of time it will rise only through the distance BA, which was crossed in the first part of time of the descent. And at the end of this last part of time the body necessarily will be found at the point A. But perhaps someone might say that it would rise higher than A, and then fall back to A. But this is absurd since in a motion due only to gravity it is not possible for a body to rise higher than the place from which it fell. For since the velocity which it has at B is again decreased by the amount BD, it is clear that, when the body is at A, it has no velocity left, and thus does not rise higher. Therefore it has been shown that the body rises to the same height from which it fell, and that each distance crossed in equal times of descent is equally measured off in as many equal times of ascent. It is also clear that in equal times equal amounts of velocity are lost. Hence the proposition is established.

In the demonstration of Proposition 2, on which the preceding proof depends, it was assumed that there exists a certain proportion between the distances crossed by a falling body in continuous equal times, and that this proportion is the same no matter what equal times are taken. This indeed must be so because of the nature of the way that things are related to each other, and if this is denied, it must be admitted that it is useless to search for a proportion between these distances. However, this proposition can be demonstrated without that assumption by using Galileo's method. Hence it will be a worthwhile effort to write down here more accurately the demonstration which he gave in a less perfect form.² Thus we will demonstrate this again.

PROPOSITION V

The distance crossed in a certain time by a body which begins its fall from rest is half the distance which it would cross in an equal time with a uniform motion having the velocity acquired at the last moment of the fall.

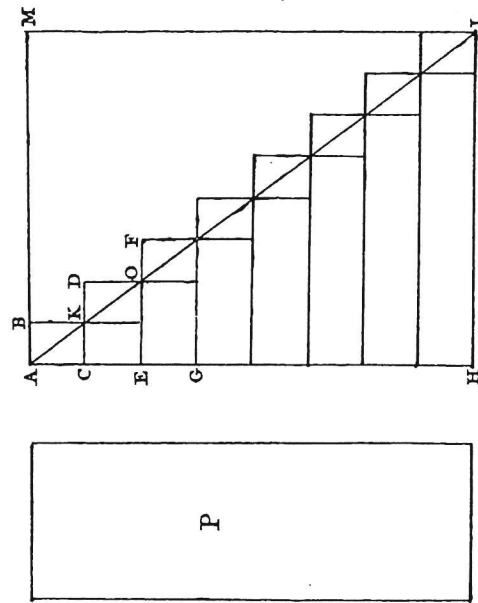
Let AH [Fig. 10] be the total time of fall. In that time the moving body crosses a distance whose quantity is designated by the plane P. Draw HL, of any length, perpendicular to AH, and let it represent the velocity acquired at the end of the fall.³ Next complete the rectangle AHLM, and let it represent the amount of distance which would be crossed in the time AH with the velocity HL. What must be shown is that the plane P is half of the rectangle MH, or, after drawing the diagonal AL, that the plane P equals the triangle AHL.

If the plane P is not equal to the triangle AHL, then it will be either smaller or larger. First assume that the plane P is smaller than the triangle AHL, if that be possible. Let AH be divided in a number of equal parts AC, CE, EG, etc. And on the triangle AHL let a figure be circumscribed which is composed of rectangles whose altitude equals each part of this division of AH, namely, the rectangles BC, DE, FG. Also within the same triangle let there be inscribed another figure composed of rectangles of the same altitude, namely, KE, OG, etc. Let all this be done in such a way that

2. Galileo, *Discourses Concerning Two New Sciences*. Third Day, Proposition 1, Theorem 1, "On Naturally Accelerated Motion" (*Le Opere di Galileo Galilei* 8:208-9). "The time in which a certain space is traversed by a moveable in uniformly accelerated movement from rest is equal to the time in which the same space would be traversed by the same moveable carried in uniform motion whose degree of speed is one-half the maximum and final degree of speed of the previous, uniformly accelerated, motion." [S. Drake's translation, 165.]

3. In the margin of his copy of the 1673 edition Huygens wrote in Latin the following, almost illegible, words: "The distance designated by the plane P should be given after the distance designated by the rectangle since HL is drawn to any length."

[Fig. 10]



the excess of the former figure over the latter figure is less than the excess of the triangle AHL over the plane P. It is clear that this can be done since the entire excess of the circumscribed figure over the inscribed figure equals the lowest rectangle, which has HL as a base. Therefore the whole excess of the triangle AHL over the inscribed figure will be less than its excess over the plane P, and thus the figure inscribed in the triangle is larger than the plane P.

Further, since the straight line AH represents the entire time of fall, its equal parts AC, CE, EG, represent equal parts of that time. And since the velocities of falling bodies increase in the same proportion as the times of fall [Proposition 1], and since HL is the velocity acquired at the end of the whole time, CK will be the velocity acquired at the end of the first part of the time, for AH is to AC as HL is to CK. Likewise EO will be the velocity acquired at the end of the second part of the time, and so forth. But it is clear that in the first time AC some distance larger than zero is crossed by the moving body. And in the second time CE the distance crossed is greater than KE since the distance KE would have been crossed in the time CE by a uniform motion with the velocity CK. For the distances crossed by a uniform motion have a ratio composed of the ratios of the times and of the velocities. And hence, since in the time AH the distance MH would have been crossed with the uniform velocity HL, it follows that in the time CE the distance KE would be crossed with the velocity CK, since the ratio of the rectangle MH to the rectangle KE is composed of the ratios of AH to CE and of HL to CK.

Now, as was said, the distance KE would be crossed in the time CE with the uniform velocity CK, but the body is moved in the time CE with an accelerated motion which already has the velocity CK at the beginning of this time. Hence it is clear that, in this accelerated motion in the time CE, a greater distance than KE is crossed. For the same reason, in the third period of time EG, a greater distance than OG is crossed since this is what would be crossed in that time EG with the uniform velocity EO. And so successively, in each part of the time AH, the body crosses a greater distance than the rectangles inscribed in the figure adjacent to each part of the time. Hence the whole distance crossed by an accelerated motion will be greater than the inscribed figure. But that distance was assumed to be equal to the plane P. Thus the inscribed figure will be smaller than the distance P, which is absurd because this was shown to be larger than that distance. Thus the plane P is not smaller than the triangle AHL. Next it will be shown that it also is not larger.

Assume that it is larger, if that be possible. Let AH be divided into equal parts, and using these parts as altitudes, let figures composed of rectangles be inscribed and circumscribed again on the triangle AHL, as before. Let this be done in such a way that the one exceeds the other by an amount less than the amount by which the plane P exceeds the triangle AHL. Hence the circumscribed figure will necessarily be less than the plane P. Now it is clear that in the first part of the time AC the distance crossed by the body is less than BC, since this would be crossed in the same time with the uniform velocity CK which the body acquires only at the end of the time AC. Likewise in the second part of the time CE the distance crossed by an accelerated motion is less than DE since this would be crossed in the same time CE with the uniform velocity EO which it acquires only at the end of the time CE. And in like fashion, in each part of the time AH, less distance is crossed by the moving body than is represented by the rectangles circumscribed on the figure adjacent to each part of the time. Hence the whole distance crossed by an accelerated motion will be less than this circumscribed figure. But that distance was assumed to be equal to the plane P. Thus the plane P will be less than the circumscribed figure, which is absurd since this figure was shown to be less than the plane P. Therefore the plane P is not larger than the triangle AHL. But it was also shown that it is not less than that triangle. Therefore it must be equal. Q.E.D.

It should be noted that everything that has been demonstrated up to this point is equally true both of bodies falling and rising on inclined planes and of bodies moved on the perpendicular. For what was said about the effect of gravity must apply to both cases for the same reason.

From this it will not be difficult to prove the following proposition which Galileo asked that we accept as in a sense being self-evident.⁴ For the demonstration which he tried to give later and which appears in the later

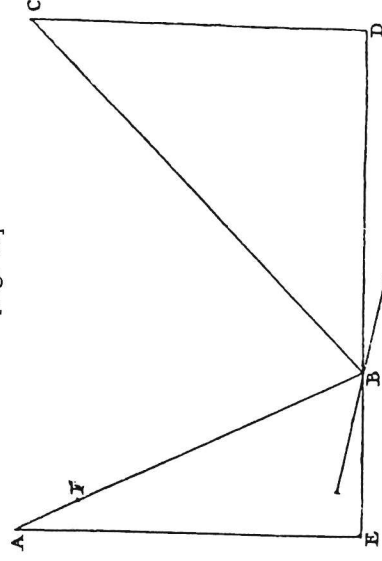
edition of his works does not seem to me to be too strong. The proposition is the following.

PROPOSITION VI

The velocities acquired by bodies falling through variably incline planes are equal if the elevations of the planes are equal.

The elevation of the plane will be called its height on the perpendicular. Let AB and CB [Fig. 11] be sections of inclined planes extended to the horizontal plane, and let their heights AE and CD be equal. Then let a body fall from A through the plane AB, and a body fall from C through the plane CB. Now I say that in each case the same degree of velocity will be acquired at the point B.

[Fig. 11]



For if we were to assume that the body falling through CB were to acquire less velocity than the body falling through AB, it would follow that the body falling through CB would acquire exactly the same velocity as a body falling only through FB, where FB is less than AB. But the body falling through CB acquires a velocity by which it could ascend again through the whole of BC [Proposition 4]. Therefore, if the body falling

4. Galileo, *Discourses Concerning Two New Sciences*. Third Day. (*Le Opere di Galileo Galilei* 8:205.) "... the Author requires and takes as true one single assumption; that is, [Postulate] I assume that the degrees of speed acquired by the same moveable over different inclinations of planes are equal whenever the heights of those planes are equal." [S. Drake's translation, 162.]

from F to B were to continue its motion through BC, which it could do by reflection in the oblique direction, it would rise up to C, that is, up to a point which is higher than the place from which it fell, which is absurd.

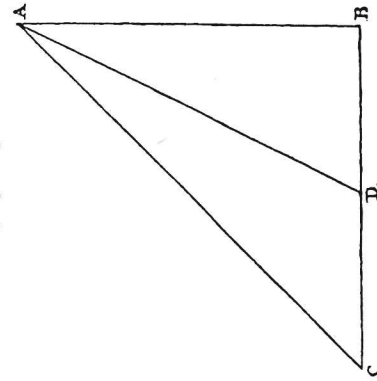
If in the place of either plane a perpendicular is substituted which has an equal height, and if a body falls through that distance, it is clear that it acquires the same velocity as the body falling on the inclined planes. For the demonstration is the same.

From this there properly follows the demonstration of another of Galileo's theorems on which are built all the other theorems which he presented concerning motions along inclined planes. This proposition is the following.

PROPOSITION VII

The times of descent on variably inclined planes whose elevations are equal are related to each other as the lengths of the planes.

Let AC and AD be inclined planes having the same elevation [Fig. 12]. Now I say that the time of descent through the plane AC is related to the time of descent through AD as the length AC is related to AD. For the time through AC is equal to the time of a uniform motion through the same AC when the velocity is half of that which is acquired by a fall through AC [Proposition 2]. Likewise the time through AD is equal to the time of a uniform motion through AD when the velocity is half of that which is acquired by a fall through AD. But these half velocities are equal to each other [Proposition 6]; thus the indicated time of a uniform motion through AC will be related to the time of a uniform motion through AD as AC to



[Fig. 12]

AD. Hence, since the times are respectively equal to these latter times, it is clear that the time of descent through AC will have the same ratio to the time of descent through AD as AC has to AD. Q.E.D.

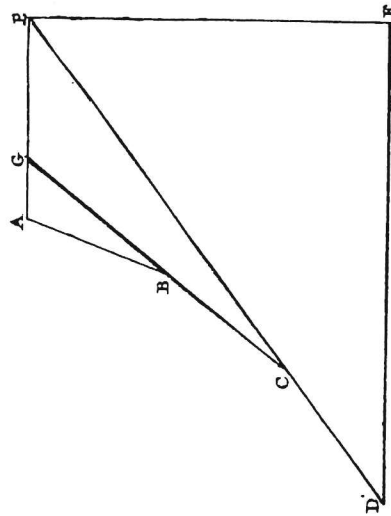
In the same way it is also shown that the time of descent through AC related to the time of fall through the perpendicular AB as AC is related to the length AB.

PROPOSITION VIII

If from the same height a body descends by a continuous motion through any number of contiguous planes having any inclinations whatsoever, it will always acquire at the end the same velocity; namely, a velocity equal to that which would be acquired by falling perpendicularly from the same height.

Let AB, BC, and CD be contiguous planes [Fig. 13] whose terminus has a height above the horizontal line DF, drawn through the lower terminus D, equal to the perpendicular EF. And let a body descend through these planes from A to D. Now I say that at D it will have the same velocity which it would have at F by falling from E.

[Fig. 13]



For when CB is extended it cuts AE at G, and likewise DC, when extended, cuts AE at E. Now a body descending through AB will acquire the same velocity at B as a body descending through GB [Proposition 6]. Hence it is clear that, if the change of direction at B has no effect, the body

will have the same velocity when it arrives at C as if it had descended through the plane GC, which is the same as it would have by descending through EC. Hence, in the same way it will cross the remaining plane CD as if it had passed through EC. And thus at D it will have the same velocity as if it had descended through the plane ED, that is, the same velocity acquired by a fall through the perpendicular EF. Q.E.D.

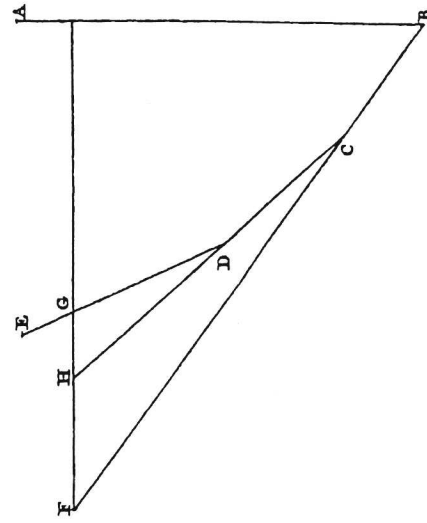
From this it also follows that a body descending through the circumference of a circle or through any curved line (for here we can consider any curve to be composed of an infinite number of straight lines) will always acquire a velocity equal to what it would acquire by descending from an equal height. This velocity will be equal to the velocity acquired in a perpendicular fall from the same height.

PROPOSITION IX

If, after falling, a body converts its motion upwards, it will rise to the same height from which it came, no matter how many contiguous plane surfaces it may have crossed, and no matter what their inclinations are.

Let a body fall from the height AB [Fig. 14]. From the point B let the planes BC, CD, and DE be inclined upwards such that their extremity E has the same height as the point A. Now I say that if a body, after falling through AB, converts its motion so that it continues to be moved through these inclined planes, it will rise up to the point E.

[Fig. 14]



For suppose, if this be possible, that the body rises only to point G. Let BC and CD be extended until they cut the horizontal GF at F and H. Now after the body has crossed the planes BC and CD, it would have only enough velocity for it to ascend through DG, or through DH, for these two velocities are equal, as was shown in Proposition 6. Hence, after having crossed the plane BC, it would only have enough velocity for it to ascend through CH, or through CF. Thus at B it has only enough velocity to ascend through BF, that is, the same velocity which it would acquire by descending through FB. But at B it has enough velocity to ascend all the way up to A. Therefore, with the velocity which it acquires by descending through FB it would be able to ascend through BA, which is higher than the place from which it descended. But this is impossible.

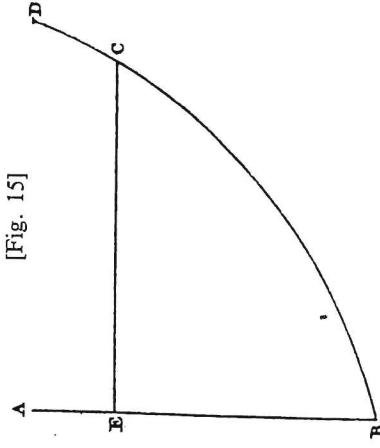
Exactly the same demonstration applies no matter how many planes are crossed by the descending body. Hence, if there be an infinite number of planes, that is, if any curved surface is given, the body will rise through it to the same height from which it came.

PROPOSITION X

If a body falls perpendicularly or through any surface, and if it later moves upwards by the acquired impetus through any other surface, then it will always have the same velocity at points of equal height in its descent and ascent.

Let a body fall from the height AB [Fig. 15] and then continue its motion through the surface BCD, in which the point C has the same height as the point E in AB. Now I say that the same velocity is present in the body at C as was present in it at E.

[Fig. 15]



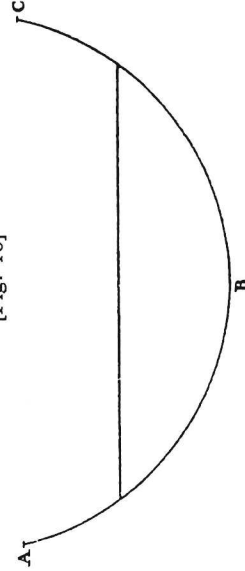
For at C it has an additional velocity by which it could ascend up to the point D, which is the same height as A [Proposition 9]. And in descending through AE it acquires a velocity by which, when the direction is reversed, it will ascend through CD [Proposition 9]. Hence it is clear that when it arrives at C in its ascent, the body has the same velocity which it had at E in its descent. Q.E.D.

PROPOSITION XI

If a body moves downwards through some surface and then reverses its motion and moves upwards through the same surface or through another similar surface similarly situated, then it will descend and ascend through the same distance in equal times.

For example, let a body descend through the surface AB [Fig. 16], and when it has arrived at B, let it reverse its motion and ascend upwards through the same surface AB or through the surface BC which is similar to AB and which is similarly situated in respect to the horizontal plane. Now it is clear from what was demonstrated earlier that it will rise to the same height from which it came. For it always has the same velocity at points of equal height in its descent and ascent [Proposition 10]. Thus it is clear that the same line is crossed twice with the same velocity in each part of the motion. Hence the times of both motions are necessarily equal. Q.E.D.

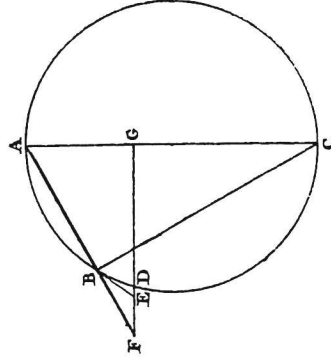
[Fig. 16]



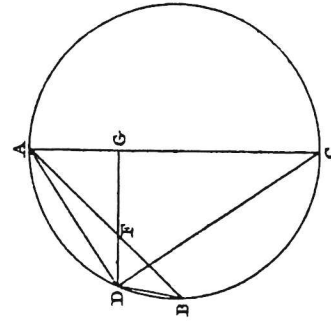
PROPOSITION XII

Let ABC [Fig. 17] be a circle having a diameter AC to which FG is drawn at right angles. Draw AF from A at the end of the diameter such that it cuts GF outside of the circle. AF must cut the circumference, for instance, at B. Now I say that the arc BD, intercepted by the lines GF and AF, is less than the straight line DF.

[Fig. 17]



[Fig. 18]



Draw the line BC, and from the point B draw the straight line BE tangent to the circumference. BE necessarily cuts the straight line FG between F and D. Now the angle BAC in the circle is equal to the angle EBC.⁵ Hence the angle FBE, which together with EBC forms the right angle FBC, will be equal to BCA. But since ABC and AGF are similar triangles, the angle F will be equal to the angle ACB. Thus this same angle F is equal to the angle FBE. Hence FEB is an isosceles triangle whose sides FE and EB are equal. If ED is added to each of these lines, then FD is equal to the sum of BE and ED. But it is clear that these latter two lines are larger than the arc BD, which is intercepted at the same two points and which is concave in the same direction. Therefore FD is larger than this same arc BD. Q.E.D.

PROPOSITION XIII

Given the same construction, if the straight line AB cuts DG inside the circle [Fig. 18], then I say that the arc BD, which is intercepted by the straight lines GD and AB, is larger than the straight line DF.

Draw the line DC, and draw DB as a cord to the arc DB. Now the angle ABD is equal to ACD, that is, to the angle ADG. And the angle DFB is larger than the angle ADF, or ADG. Hence the same angle DFB is also larger than DBF. Hence in the triangle DFB the side DB is larger than the side DF. And therefore the arc DB is even still larger than the line DF. Q.E.D.

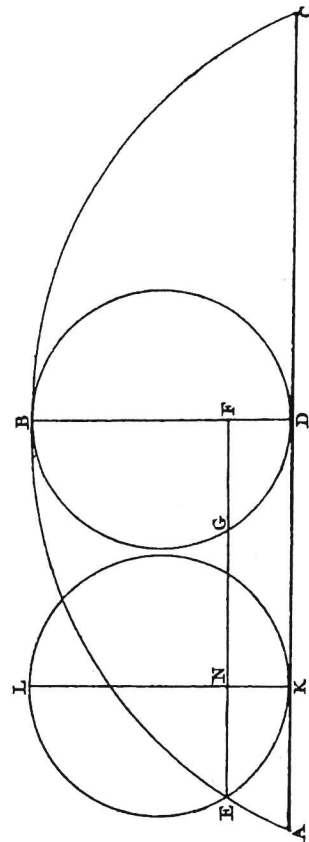
⁵ 's Gravesande has noted that this is Proposition 32, Book 3, of Euclid's *Elements*. "If a straight line touch a circle, and from the point of contact there be drawn across, in the circle, a straight line cutting the circle, the angles which it makes with the tangent will be equal to the angles in the alternate segments of the circle." [T. L. Heath's translation 2:65.]

Since an asterisk for a reference appears in the text of the 1673 edition at this point, Huygens no doubt intended to include this reference to Euclid.

PROPOSITION XIV

Let ABC [Fig. 19] be a cycloid whose base is AC and whose axis is BD . I think that it is clear how this figure is generated from what was said above⁶ about the definition and the mechanical description of the cycloid. Around its axis BD let the circle BGD be drawn. From some point E on the cycloid draw EF parallel to the base AC . EF cuts the axis BD at F and the circumference BGD at G . Now I say that the straight line GE is equal to the arc GB .⁷

[Fig. 19]



Through the point E draw the circle LEK which is equal to BGD and which is tangent to the base of the cycloid at K . Draw the diameter KL . Now the straight line AK is equal to the arc EK , and the whole line AD is equal to the semicircumference KEL . Therefore, KD is equal to the arc EL or GB . But KD or NF is equal to EG , since EN equals GF , and NF is common to both. Thus it is proven that GE is equal to the arc GB .

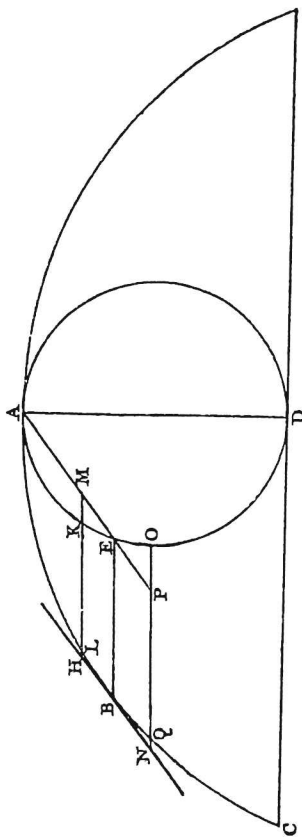
PROPOSITION XV

Given a point on a cycloid, through that point to draw a straight line which is tangent to the cycloid.

Let ABC [Fig. 20] be a cycloid, and let B be the given point through which the tangent must be drawn.

Around the axis AD of the cycloid draw the generating circle AED .

[Fig. 20]



Draw BE which is parallel to the base of the cycloid and which cuts the indicated circle at E . Draw AE , and finally through B draw HBN parallel to AE . Now I say that HBN is tangent to the cycloid at B .

On the line HBN take any point other than B , and first take some point H above B . Through H draw a straight line parallel to the base which cuts the cycloid at L , the circle AED at K , and the line AE at M . Now KL is equal to the arc KA , and the straight line KM is less than the arc KE . Hence the straight line ML will be less than the arc AE , that is, less than the line EB , or MH . From this it is clear that point H lies outside the cycloid.

Next on the line HN take some point N below B . Through N draw, as before, a parallel to the base, which cuts the cycloid at Q , the circle AED at O , and the extended straight line AE at P . Now OQ is equal to the arc OA , but OP is larger than the arc OE . Hence PQ will be less than the arc EA , that is, less than the straight line EB , or PN . From this it is clear again that the point N lies outside the cycloid. Therefore, since every point except B on the straight line HBN lies outside the cycloid, it is proven that HBN touches the cycloid at the point B . $Q.E.D.$

I had some doubts as to whether I should provide a special place for this demonstration since I have found that a very similar proof by Mr. Wren has been published in a book by Mr. Wallis⁸ on the cycloid. However, it is possible to establish the proposition by using a universal method which applies not only to cycloids but to other curves generated by the rolling of a figure, as long as that figure is concave in the same direction and is one of those figures which are called geometrical.

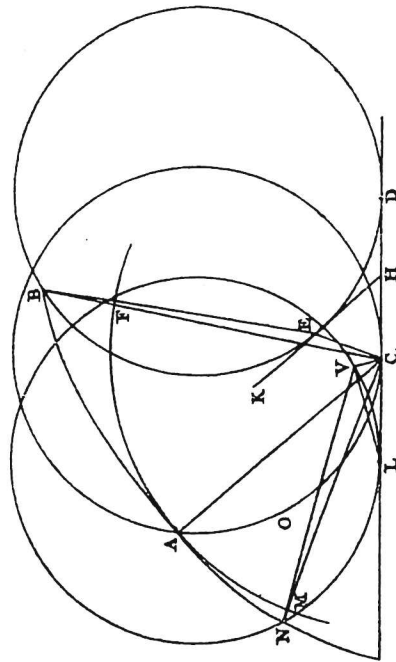
Let NAB [Fig. 21] be a curve generated by the rolling of the figure OL on the line LD and traced out by the point N taken in the circumference of

6. Part I, paragraphs 12ff.

7. Huygens discovered this proposition in 1658. See *Oeuvres* 14:347.

8. John Wallis, *Tractatus duo, prior de cycloide et corporibus inde genitis; posterior, epistolaris in quo agitur de cissoide* . . . (Oxoniae: Typ. Ac. Lichfieldianis, 1659). "De recta tangente cycloidem primariam," 71.

[Fig. 21]



the figure OL.⁹ One could draw a tangent at point A of the curve. Draw the straight line CA from the point C where the figure touched the line LD when the generating point was at A. This point of contact can always be found since the problem reduces to drawing two straight parallel lines, of which one is tangent to the generated figure and the other passes through the generating point given in the rotating figure such that these parallels are separated from each other by the same distance by which the point A is separated from the line LD. Now I say that CA cuts the curve at right angles; in other words, the circumference MAF described by the radius CA around the center C touches the curve at the point A; hence a perpendicular to AC drawn through the point A is tangent to the curve at this point.

First draw CB to the point B of the curve which is located further above the line LD than the point A. Consider the figure to be located at BED where the generating point would be at B, the point of contact with LD at D, and the point of the curve which was at C when the generating point was at A has now risen to E. Draw EC and EB, and draw a tangent to the figure at E which cuts LD at H.

Now the straight line CD is equal to the curve ED. And the sum of EH and HD is larger than this curve. Thus EH is larger than CH. Hence the angle ECH is larger than CEH, and thus ECL is smaller than CEK. By adding the angle KEB, which is equal to LCA, to KEC, we get the angle CEB. And by subtracting the angle LCB from ECL, we get ECB. Hence the angle CEB is very much larger than the angle ECB. Thus in the triangle CEB, the side CB is larger than EB. But EB is clearly equal to CA since this

9. Huygens here develops an idea borrowed from Descartes and van Schooten. See *Oeuvres* 18:388–

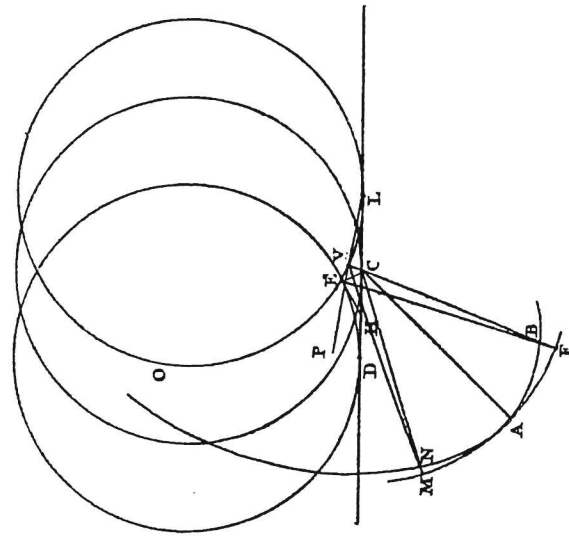
is the same line which was rotated with the figure. Hence CB is also larger than CA, that is, larger than CF. Therefore it is established that the point I lies outside of the circumference MAF.

Next take the point N located on the curve between LD and the point A. When the generating point is at N, let the figure be situated at VL, the point of contact at L, and the point which previously touched LD at C has now risen to V. Draw CN, NV, VC, and VL. Now VN equals CA since it is identical to CA rotated to VN. And the straight line LC is equal to the curve LV, and thus is larger than the straight line LV. Hence in the triangle CLV the angle LVC is larger than LCV. If we add the upper angle LVN to LVC, we get the whole angle NVC which is larger than LCV, and which is thus very much larger than the angle NCV which is part of LCV. Therefore in the triangle CVN the side CN is larger than the side VN, which is equal to CA. Thus CN is larger than CA, that is, larger than CM. From this it is apparent that the point N lies outside the circle MAF, which touches the curve at the point A. Q.E.D.

The construction and the demonstration are the same whether the curve is produced by a generating point located either inside or outside the boundary of the rotating figure. But in the latter case part of the curve descends below LD, and because of this some differences arise in the demonstration.

Let the point A, through which a tangent could be drawn, be taken on the curve NAB which descends below the line CL [Fig. 22]. NAB is described by a point N located outside the rotating figure, but it has

[Fig. 22]



definite position in the same plane. Having found the point C where the rotating figure touches CD when the generating point was at A, draw the straight line CA. Now I say that CA cuts the curve NAB at right angles; in other words, the circumference described by the radius CA around the center C is tangent to the curve NAB at the point A. But it will be shown that this circumference touches the curve from the outside, whereas in the part of the curve located above the line CD, it touches it from the inside.

All things being given and described as before, it is clear again that the angle ECH is larger than CEH. By adding HCB to ECH we get the angle ECB, and by subtracting HEB, which is equal to DCA, from CEH, we get the angle CEB. Hence ECB is much larger than CEB. Thus in the triangle ECB the side EB is larger than CB. But EB is equal to CA or CF. Hence CF is also larger than CB. Therefore the point F of the circumference lies outside the curve as compared to the center.

Next it is clear that the angle LVC is larger than LCV. Thus CVP, which together with LVC equals two right angles, is less than VCD. By adding angle DCN to VCD, we get VCN, and by subtracting angle PVN from CVP, we get CVN. Hence the angle VCN is much larger than CVN. Thus in the triangle CVN the side VN is larger than CN. But VN is equal to CA or CM. Thus CM is also larger than CN, and therefore the point M of the circumference lies outside the curve NAB as compared to the center. Thus it is established that the circumference touches the curve at point A. Q.E.D.

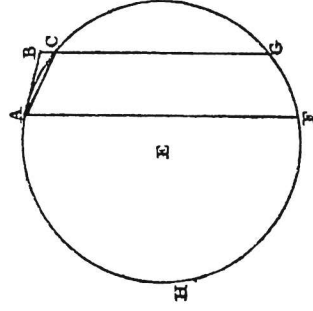
Further, if the point of the curve through which the tangent is to be drawn is the same point at which the radius cuts the curve, then the sought-for tangent will always be perpendicular to the radius, as is easily shown.

PROPOSITION XVI

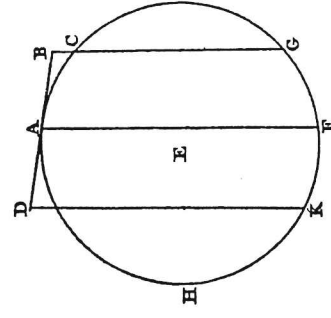
Let AF and BG [Fig. 23] be two straight parallel lines which cut the circumference of a circle whose center is E. Let both of these lines be located on the same side of the center, or let one of them, AF, pass through the center. Draw a tangent at point A which is located on the line cutting the circumference which is closer to the center. Now I say that the part AB of the tangent intercepted by these two parallels is less than the arc AC intercepted by the same two parallels.

Draw the cord AC on the arc AC. Now the angle BAF is equal to the angle contained by the portion AHF of the circle, which is larger than or equal to a semicircle. Hence the angle BAF is less than or equal to a right angle. Thus the angle ABC is larger than or equal to a right angle. Therefore in the triangle ABC the side AC under the angle B is larger than the

[Fig. 23]



[Fig. 24]



side AB. But this side AC is smaller than the arc AC. Hence AB is clearly smaller than the arc AC.

PROPOSITION XVII

Given the same circumstances, let there be a third parallel line DI [Fig. 24] which cuts the circle and which is the same distance from AF which was closer to the center, as AF is from BG. Now I say that the part of the tangent at A intercepted by this added parallel and the middle parallel, that is, the line AD, is less than the arc AC which is intercepted by the first two parallels.

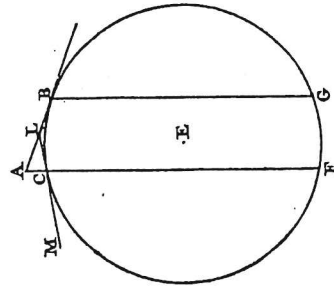
This is obvious since AD is equal to AB which we have already shown: is less than the arc AC.

PROPOSITION XVIII

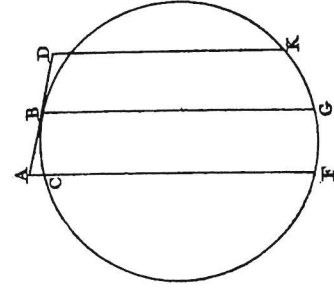
Let AF and BG [Fig. 25] be two parallel lines which cut through a circle whose center is E. Draw a straight line tangent to the circumference at B which is the point where the circumference is cut by the parallel which is either further removed from the center, or the same distance from the center, than the other parallel. Now the part BA of this tangent intercepted by the parallels is larger than the arc BC intercepted by the same parallels.

At the point C draw the tangent MCL to the circumference. This cut the tangent BA at L. In the triangle ACL the angle C is equal to the angle MCF, that is, to the angle contained in the portion CBF of the circle. Bu

[Fig. 25]



[Fig. 26]



the angle A is equal to the angle contained by the portion BCG of the circle, which portion is larger than or equal to the portion CBG since BG is either more distance from the center, or the same distance from the center, than CF. Thus in triangle ACL the angle A is either less than or equal to the angle C. Hence the side CL is either less than or equal to the side AL. But CL plus LB is larger than the arc CB. Therefore AL plus LB, that is, the tangent AB, will be larger than the arc CB. Q.E.D.

PROPOSITION XIX

Given the same circumstances, let there be a third straight line DK [Fig. 26] which is parallel to the first two, which cuts the circle, and which is the same distance from the parallel which is further from the center as the latter is from the other parallel AF. Now the part of the tangent at B intercepted by the middle parallel and the added line DK, that is, the line BD, will be larger than the arc BC.

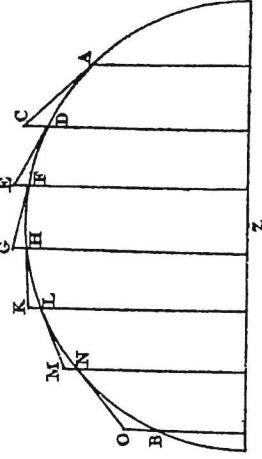
This is obvious since BD is equal to BA which we have already shown is larger than the arc BC.

PROPOSITION XX

Let AB [Fig. 27] be an arc of a circle which is less than a semicircumference and which is cut into any number of parts by parallel straight lines. Let there be an equal distance between the parallels and between the mutually parallel lines drawn through the ends of the arcs; namely, CD, EF, GH, KL, etc. Draw tangents to the circumference in the same direction at

the point A at the end of one arc and at all the other points of the other sections, and let each tangent cut the next parallel line, as do the tangents AC, DE, FG, HK, etc. Now I say that these tangents taken together, but minus the first one AC, are less than the given arc AB; and that these same tangents, including AC, are greater than the arc AB decreased by its last part NB, that is, greater than the arc AN.

[Fig. 27]



First let us grant that some of the parallels are located on both sides of the center Z. Among those located on the side B, let GH be either the closest to the center or let it pass through the center. Thus each of the tangents located between GH and BO, that is, HK, LM, and NO, is less than its corresponding arc [Proposition 16]. Also the tangent GF is less than the next arc FD [Proposition 17], and likewise the tangent ED is less than the arc DA. Thus all the tangents located between BO and CD are less than the arcs BH plus FA, and hence are clearly less than the arcs BH plus HA, or less than the arc BA. Q.E.D.

Next we will prove that all the tangents between BO and A are larger than the arc AN. For the parallel GH will be either closer to the center Z than is the parallel EF, which I assume is the closest to the center of the parallels on the side of A, or it is further away, or it is equidistant.

Now if EF is further from the center than GH, or is equidistant, then the tangent FG will be larger than its arc FH, and the remaining tangents on the side A, that is, ED and CA, will also each be larger than their arcs [Proposition 18]. And thus the sum of GF, ED, and CA is larger than the arc HA. But the tangent LM will be larger than the arc LN, and thus the sum of all the tangents, except HK, will be larger than the arc AN, and thus the sum of all the tangents, when HK is added, all the tangents located between A and B will be larger than the arc AN.

But if GH is further from the center than EF, then the tangent KH will

be larger than the arc HF [Proposition 19], the tangent ML as before is larger than the arc LH, and the tangent ON is larger than the arc HL; and thus the sum of the tangents ON, ML and KH is larger than the arc NF. And the tangent ED is larger than its own arc FD [Proposition 18], and the tangent CA likewise is larger than its arc DA. Thus the sum of the tangents between BO and A, excepting GF, is larger than the arc NA. Thus clearly, when GF is added, these tangents, that is, all which are located between BO and A, will be larger than the arc NA.

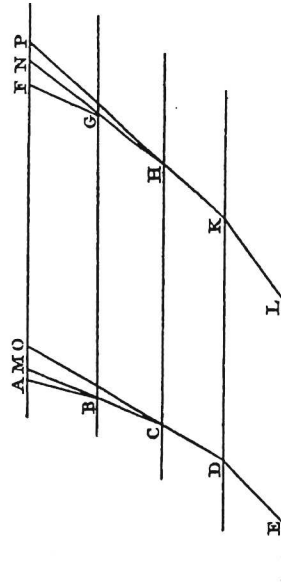
From this the demonstration in other cases is clear, no matter what arc of a semicircumference is taken, for it will be either the same as, or only a part of, the preceding demonstration.

PROPOSITION XXI

Let a body descend by a continuous motion through any number of contiguous planes, and later let it descend from the same height through another series of an equal number of contiguous planes. Let the latter series be constructed in such a way that each plane corresponds in height to another plane in the first series, but let the planes in the second series have a larger inclination than those in the first series. Now I say that the time of descent through the less inclined planes will be less than the time of descent through the more inclined planes.

Let there be two series of planes ABCDE and FGHKL located between the same horizontal parallels [Fig. 28]. Let these planes correspond to each other two by two, and let the planes of each series be contained within the same horizontal parallels. Also let each plane of the series FGHKL be more inclined to the horizontal than the plane of corresponding height in the series ABCDE. Now I say that the descent through ABCDE will occur in a shorter time than the descent through FGHKL.

[Fig. 28]



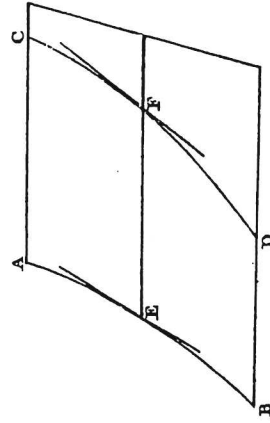
First it is clear that the time of descent through AB is shorter than the time of descent through FG, since the ratio of these times is equal to the ratio of AB to FG [Proposition 7], and AB is less than FG because of its smaller inclination. Next let the lines CB and HG be extended upwards, and let them cut the horizontal AF at M and N. Now the time through BC after AB is equal to the time through this same BC after MB because at the point B the body has the same velocity whether it descended through AB or through MB [Proposition 6]; and likewise the time through GH after FG is equal to the time through this same GH after NG. But the time through BC after MB is related to the time through GH after NG as BC is related to GH in length, or as CM to HN. For the same ratio is found between the times through the whole lengths MC and NH, and through the partial lengths MB and NG [Proposition 7], and hence also between the remaining times. But BC is less than GH because of its smaller inclination. Therefore it is clear that the time through BC after MB or after AB is shorter than the time through GH after NG or after FG.

By extending DC and KH upwards until they cut the horizontal AF at O and P, it is proven in the same way that the time through CD after ABC or after OC is shorter than the time through HK after FGH or after PH. And finally the time through DE after ABCD is shorter than the time through KL after FGHK. Therefore the whole time of descent through ABCDE will be shorter than the time of descent through FGHKL. Q.E.D.

Next, if we consider curved lines to be composed of innumerable straight lines, then from the above it is clear that if there be two inclined surfaces which correspond to curved lines of the same height, and if at every point of the same height the inclination of the one surface is always larger than the other, then a body will fall through the less inclined surface in a shorter time than through the greater inclined surface.

For example, let there be two inclined surfaces [Fig. 29] corresponding to the curves AB and CD, which have the same height, and which are such that at any points E and F of equal height CD is more inclined than AB,

[Fig. 29]



that is, the tangent to the curve at F is more inclined to the horizontal than the tangent to the curve at E. Then the time of descent through AB will be shorter than through CD.

The same thing occurs if one of the lines is a straight line, provided that the inclination of the straight line, which is the same everywhere, is larger or smaller than the inclination of the curve at every one of its points.

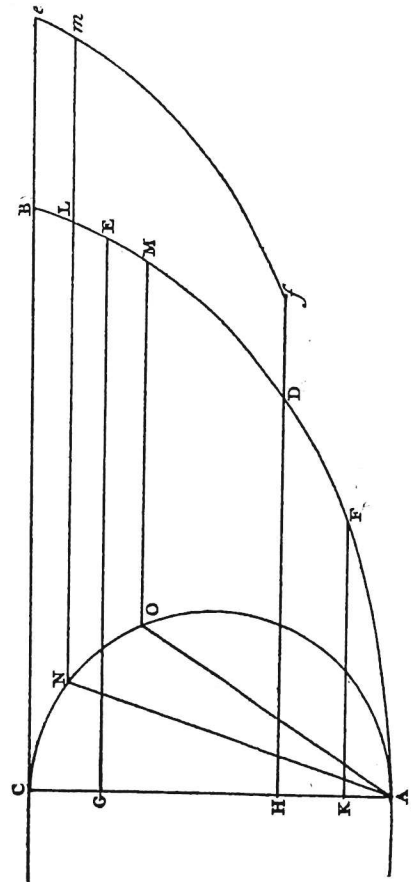
PROPOSITION XXII

Let there be a cycloid whose axis is erected as a perpendicular with its vertex located at the bottom. And let there be taken two portions of the curve having equal height but with the one being closer than the other to the vertex. Then the time of descent through the upper portion of the curve will be shorter than the time through the lower portion.

Let AB be a cycloid whose axis AC is erected on the perpendicular and whose vertex A is located at the bottom [Fig. 30]. On the cycloid mark off the portions BD and EF having equal heights, that is, the parallel horizontals BC and DH, which contain the upper portion BD, have the same distance from each other as the parallels EG and FK, which contain the lower portion EF. Now I say that the time of descent through the curve BD will be shorter than the time through EF.

Take any point L on BD, and any point M on EF, such that E has the same height above M as B has above L. After drawing a semicircle around the axis AC, the horizontal lines LN and MO will cut the semicircle at N and O. Draw NA and OA. Now since N is higher than O, it is clear that the

[Fig. 30]

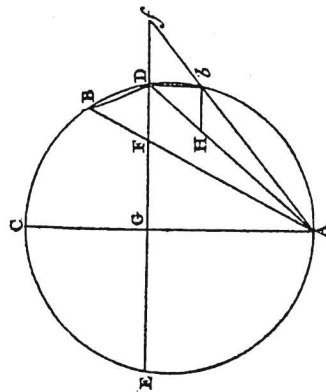


line NA is less inclined to the horizontal than is OA. But the tangent to the curve at L is parallel to NA [Proposition 15], and the tangent to the curve at M is parallel to OA. Therefore the curve BD at the point L is less inclined than the curve EF at the point M. Now let the portion EF, its inclination remaining invariant, be understood to be moved higher to the position *ef* such that it is contained between the same parallels as the portion BD. Then the point M will be found at *m*, at a height equal to the point L. Therefore the inclination of the curve *ef* at the point *m*, which is the same as the inclination of the curve BD at L. In the same way it is shown that, at any other point of the curve *ef*, the inclination is greater than the inclination at an equally high point in the curve BD. Hence the time of descent through BD will be shorter than the time through *ef* [Proposition 21], or which is the same thing, through EF. Q.E.D.

LEMMA

Let AC [Fig. 31] be the diameter of a circle which cuts the line DE at right angles. From the end A of the diameter draw the line AB which cuts the circumference at B and the line DE at F. Now I say that these three lines, AB, AD, and AF, are proportional.

[Fig. 31]



First assume that the point of intersection F lies within the circle; and draw the cord corresponding to the arc BD. Since the arcs AE and AD are equal, the angles EDA and ABD, which are contained in these portions of the circumference, are equal. Thus in the triangles ABD and ADF, the angles ABD and ADF are equal. But each has a common angle at A. Hence

the indicated triangles will be similar, and therefore BA is related to AD as AD is to AF.

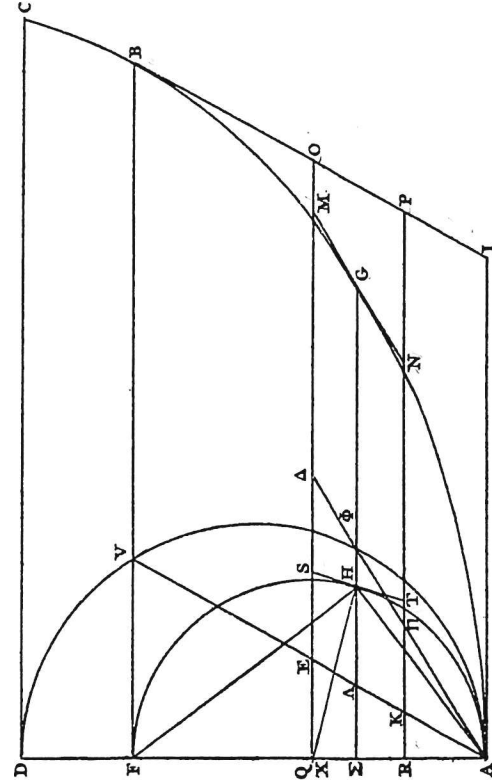
But if the point of intersection f lies outside the circle, draw bH parallel to DE. This line cuts AD as H. Thus, according to what was just demonstrated, DA will be related to Ab as Ab is to AH, that is, as Af to AD. Therefore again the lines Af, AD, and Ab will be proportional. Thus the proposition is established.

PROPOSITION XXIII

Let ABC [Fig. 32] be a cycloid whose vertex A is located at the bottom and whose axis AD is erected on the perpendicular. Select any point B on the cycloid, and draw from it downwards the line BI which is tangent to the cycloid and which ends at the horizontal line AI. Draw BF perpendicular to the axis. Divide FA in half at X and draw on it the semicircle FHA. Next through any point G on the curve BA draw the line ΣG parallel to BF. This line cuts the circumference FHA at H and the axis AD at Σ . Through the points G and H imagine that tangents are drawn to each curve, and let MN and ST be parts of these tangents intercepted by the two horizontal lines MS and NT. Between these same lines MS and NT are contained the part OP of the tangent BI and the part QR of the axis DA.

Given this construction, I say that the time in which a body crosses the line MN, with the uniform velocity acquired after it has fallen through the

[Fig. 32]



arc BG of the cycloid, will be related to the time in which it would cross the line OP, with half of the uniform velocity which it would acquire by falling through the whole tangent BI, as the tangent ST is related to the part QR of the axis.

On the axis AD draw the semicircle DVA, which cuts BF at V and ΣG at Φ . Draw AV which cuts OQ at E, PR at K, and G Σ at Λ . Draw HF, HA, HX, and A Φ , the latter of which cuts OQ at Δ and PR at Π .

The previously mentioned time through MN is related to the time through OP by a ratio composed of the ratio of these same lines MN to OP plus the inverse ratio of the velocities by which they are crossed,¹⁰ this latter being the ratio of half of the velocity from BI or from FA to the velocity from BG or from F Σ [Proposition 8]. But the whole velocity from FA is related to the velocity from F Σ as the ratio of the square roots of the lengths FA to F Σ [Proposition 3], and thus the same as FA to FH. Therefore half the velocity from FA will be related to the velocity from F Σ as FX to FH. Thus the indicated time through MN is related to the time through OP by a ratio composed of the ratios of MN to OP and of FX to FH. But we will show that the first of these ratios, that is, MN to OP, is the same as FH to H Σ .

For the tangent BI of the cycloid is parallel to VA, and likewise the tangent MGN is parallel to ΦA . And thus MN equals $\Delta \Pi$, and OP equals EK. Therefore the indicated ratio of MN to OP is the same as $\Delta \Pi$ to EK; that is, as ΔA to EA; that is, as ΦA to ΛA ; that is, as VA to ΦA [preceding Lemma]. But VA is to A Φ as FA is to AH. For since the square of VA equals the rectangle DAF, and since the square of A Φ equals the rectangle D $\Lambda \Sigma$, which rectangles are related to each other as FA to ΣA , that is, as the square of FA to the square of AH, then from this the square of VA is to the square of ΦA as the square of FA is to the square of AH, and also VA is to A Φ in length as FA is to AH. Thus the ratio of MN to OP is equal to the ratio of FA to AH, that is, it is the same as the ratio of FH to H Σ , as was said, because FAH and FH Σ are similar triangles. Thus the indicated ratio of time through MN to the time through OP is composed of the ratios of FX to FH and of FH to H Σ , and therefore will be the same as the ratio of FX or XH to H Σ . But the radius XH is to H Σ as the tangent ST is to the line QR, as is easily seen. Therefore, it is clear that the time of motion which we mentioned through MN is related to the time through OP as ST to QR. Q.E.D.

10. In the 1673 edition Huygens included a reference at this point to Galileo, *Discourses Concerning Two New Sciences*, Third Day, Proposition 5, "On Equable Motion" (*Le Opere di Galileo Galilei* 8:195). "If two moveables are carried in equable motion but with unequal speeds, and unequal spaces are run through, then the ratio of the times will be compounded from the ratio of the spaces and from the inverse ratio of speeds," [S. Drake's translation, 151.]

the velocity from BT is related to the time through the line ΔE , with half the velocity from FA, as the tangent $\Gamma\Sigma$ is related to the line PQ. And so forth, each of the times through tangents of the cycloid, which are the same as mentioned above, will be related to the times of uniform motion through the corresponding parts of the line BI, with half the velocity from B Θ , as the tangents to the circumference FH, contained between the same parallels, are related to the corresponding parts of the line FG.

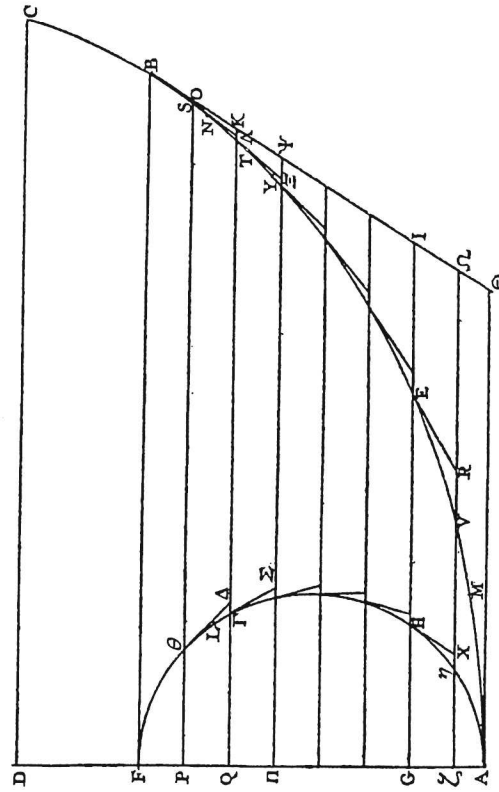
Therefore, we have certain quantities in the straight lines FP, PQ, etc., and an equal number of other quantities in the times in which the lines BA, ΔE , etc., are crossed with a uniform motion with half the velocity from B Θ . And each quantity in the former series is related to its successor in the same proportion that each quantity of the second series is related to its successor. For the quantities in each series are equal to each other. Furthermore the quantities in the first series are related to other quantities; namely, to the tangents to the circle ΔX , $\Gamma\Sigma$, etc., in the same proportions and in the same order as the quantities in the second series are related to other quantities; namely, to the times of the indicated motions through the tangents of the cycloid VS, MT, etc. Therefore, the sum of all the first quantities is related to the sum of the quantities to which they are related, that is, the whole line FG is related to all the tangents XA, $\Gamma\Sigma$, etc. as the time in which the whole of BI is crossed, with half the velocity from B Θ , will be related to the sum of the times of all the indicated motions through the tangents of the cycloid VS, MT, etc.¹¹ Thus, by inversion, the ratio of the times of the indicated motions through the tangents of the cycloid to the time through the line BI, with half the velocity from B Θ , will be the same as the ratio of all the indicated tangents to the circumference FH to the straight line FG. But this ratio is less than the ratio of the arc FO to the same line FG, because the arc F Φ , and thus also the arc FO, is larger than all of the indicated tangents to the arc FH [Proposition 20]. But the time through BE after NB was assumed to be related to the time through BI, with half the velocity from B Θ , as the arc FO is related to the straight line FG. Therefore the sum of the indicated times through the tangents of the cycloid will be less than the time through BE after NB, while previously it was shown that it is larger. This is absurd. Therefore the time through the arc BE of the cycloid has a ratio to the time through the tangent BI, with half the velocity from B Θ or from FA, which is not larger than the ratio of the arc FH of the circumference to the straight line FG.

Next, assume that this ratio is smaller, if that be possible. If so, there will be some time greater than the time through the arc BE (let this time be

Z) which will be related to the indicated time through BI as the arc FH related to the straight line FG.

Let the arc NM [Fig. 34] be equal in height to the arc BE, and let its upper terminus point N be lower than the point B. Then the time through the arc NM will be shorter than the time through the arc BE [Proposition 22]. Now it is clear that the point N can be taken so close to the point B that the difference between these times is as small as one wishes, and hence would be less than the time by which Z exceeds the time through the arc BE. Let the point N be so chosen. On this basis the time through NM will be shorter than the time Z, and will have a ratio to the indicated time through BI, with half the velocity from B Θ , which is less than the ratio of the arc FH to the straight line FG. Hence this ratio would be equal to the ratio of the arc LH to the straight line FG.

[Fig. 34]



Divide FG into equal parts FP, PQ, etc., each of which is less in height than the arc BN of the cycloid and also less in height than the arc FL of the circumference. After adding to FG one more of these parts G ζ , draw from the points of division the lines PO, QK, etc., which are parallel to the base DC and which terminate at the tangent B Θ . Next from the point ζ draw the straight line ζQ which cuts the cycloid at V and the circumference at η . From each of the points where the drawn parallels cut the circumference FH, draw tangents downwards as far as the next parallel, for example, $\Theta\Delta$, $\Gamma\Sigma$. From the lowest of these points H draw the tangent which cuts the line

11. In the 1673 edition Huygens included a reference at this point to Proposition 2 of Archimedes' *On Spheroids and Conoids*. See T. L. Heath, ed. *The Works of Archimedes* (New York: Dover, n.d.) 110.

$\xi\Omega$ at X. Likewise, from the points where the indicated parallels cut the cycloid, draw the same number of tangents downwards, for example, SA, T Ξ . The lowest of these, namely, the tangent drawn from the point E, cuts the line $\xi\Omega$ at R.

Now P ξ is equal to FG, which is the height of the arc BE. And the altitude of the arc NM is equal to that of the arc BE from the construction. Thus P ξ will be equal to the altitude of the arc NM. But from the construction the line PO is above the terminal point N. Therefore, the line $\xi\Omega$, and in it the point V, is above the terminal point M. Hence, since the arc SV has the same height as the arc NM but its terminus S is higher than N, the time through SV will be shorter than the time through NM [Proposition 22].

But the time through the tangent SA with the uniform velocity from BS is shorter than the time of an accelerated fall through the arc ST beginning at S. For the velocity from BS, by which it is assumed that the whole of SA is crossed, is equal to the velocity from ST [Proposition 8], which is acquired at the end of a motion through the arc ST; and SA is less than ST. Likewise the time through the tangent T Ξ , with the uniform velocity from BT, is shorter than the time of an accelerated fall through the arc TY after ST. For the velocity from BT, by which it is assumed that the whole of T Ξ is crossed, is equal to the velocity from SY, which is acquired at the end of the indicated motion through the arc TY after ST, and T Ξ is less than the arc TY. And thus the sum of the times of the uniform motions through the tangents of the cycloid, with the velocity through each being equal to the velocity acquired by a fall from B up to that point of contact, will be shorter than the time of an accelerated fall through the arc SV. But this same sum will also be longer, as we will now show.

The indicated time through the tangent SA with the uniform velocity from BS is related to the time through the line OK, with half the uniform velocity from B Θ , as the tangent $\Theta\Delta$ of the semicircle is related to the line PQ [Proposition 23]. Likewise the time through the tangent T Ξ with the uniform velocity from BT is related to the time through the line K Ψ , with half the uniform velocity from B Θ , as the tangent $\Gamma\Sigma$ is related to the line QII. And thus successively each of the times through the tangents of the cycloid, which are the same as mentioned above, will be related to the times of uniform motion through the corresponding parts of the line O Ω , with half the velocity from B Θ , as the tangents to the circumference $\Theta\eta$, contained within these same parallels, are related to the corresponding parts of the line P ξ . Hence, as in the first part of this demonstration, we conclude that the sum of the lines PQ, QII, etc., that is, the whole line P ξ , is related to the sum of the tangents $\Theta\Delta$, $\Gamma\Sigma$, etc., as the time in which the whole of O Ω is crossed, with half the velocity from B Θ , is related to the sum of the times of the indicated motions through the tangents of the cycloid SA, T Ξ , etc. And thus, by conversion, the sum of the times through the tangents to

the cycloid will have the same ratio to the indicated time of uniform motion through the line O Ω , or through the line BI, as the sum of the indicated tangents to the arc $\Theta\eta$ has to the straight line P ξ or FG. As a result the ratio of the times is larger than the ratio of the arc LH to the straight line FG. For the arc ΘH , and thus clearly also the arc LH, is less than the indicated tangents of the arc $\Theta\eta$ [Proposition 20]. But we postulated at the beginning that the time through NM has the same ratio to the time through BI as the arc LH has to the straight line FG. Therefore the time through NM, and much more the time through SV, will be less than the time through the tangents to the cycloid. But this is absurd, since this time was shown above to be less than the time through the arc SV. Therefore it is clear that the ratio of the time through the arc BE of the cycloid to the time through the tangent BI, with half the uniform velocity from B Θ , is not less than the ratio of the arc FH to the straight line FG.

But it was also shown that this ratio is not larger. Therefore it must be the same. Q.E.D.

PROPOSITION XXV

On a cycloid whose axis is erected on the perpendicular and whose vertex is located at the bottom, the times of descent, in which a body arrives at the lowest point at the vertex after having departed from any point of the cycloid, are equal to each other; and these times are related to the time of a perpendicular fall through the whole axis of the cycloid with the same ratio by which the semicircumference of a circle is related to its diameter.

Let ABC [Fig. 35] be a cycloid whose vertex A is located at the bottom and whose axis AD is erected on the perpendicular. Select any point on the cycloid, for example B, and let a body descend by its natural impetus through the arc BA, or through a surface so curved. Now I say that the time of this descent is related to the time of a fall through the axis DA as the semicircumference of a circle is related to its diameter. When this has been demonstrated, it will also be established that the times of descent through all arcs of the cycloid terminating at A are equal to each other.

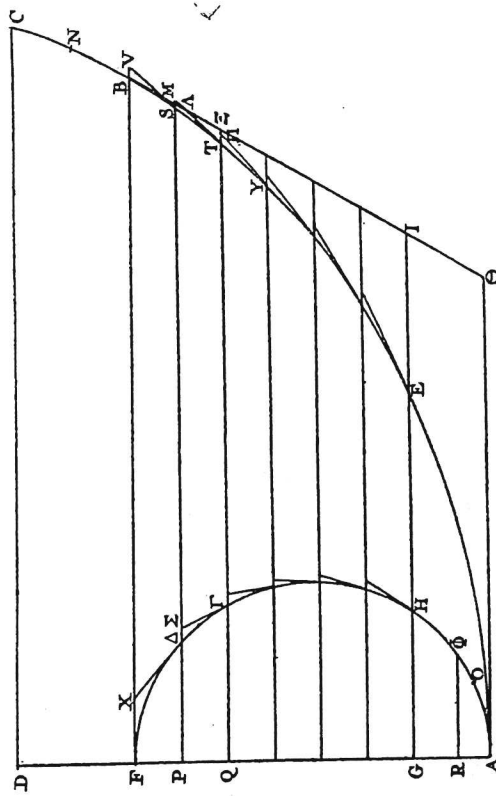
On the axis DA draw a semicircle whose circumference cuts the straight line BF, which is parallel to the base DC, at E. Draw the line EA, and parallel to it draw the line BG which is tangent to the cycloid at B. BC will cut the horizontal straight line drawn through A at G. Also on FA draw the semicircle FHA.

Now according to the preceding proposition, the time of descent through the arc BA of the cycloid is related to the time of a uniform motion through the line BG, with half the velocity from BG, as the semicircular arc

PROPOSITION XXIV

Again as in the preceding proposition, let ABC [Fig. 33] be a cycloid whose vertex A is located at the bottom and whose axis AD is erected on the perpendicular. Having taken any point B on the cycloid, draw from it downwards the straight line $B\Theta$ which is a tangent to the cycloid and which cuts the horizontal AO at Θ . Draw BF perpendicular to the axis, and around FA draw the semicircle FHA . Finally draw another straight line GE , parallel to FB , which cuts the cycloid at E , the line $B\Theta$ at I , the circumference FHA at H , and the axis DA at G .

[Fig. 33]



Now I say that the time of descent through the arc BE of the cycloid is related to the time of descent through the tangent BI with half the velocity acquired from a fall through $B\Theta$ as the arc FH is related to the straight line FG .

For if this is not true, then the ratio of the time through the arc BE to the indicated time through BI will be either greater or smaller than the ratio of the arc FH to the straight line FG . Let us assume first that it is greater, if that be possible.

If so, then there will be some time shorter than the time through BE (let this be time Z) which will be related to the indicated time through BI as the arc FH is related to the straight line FG . Now if we take some other point N above B on the cycloid, then the time through BE after NB will be

shorter than the time through BE . But it is clear that the point N can be taken so close to B that the difference of these times is as small as one wishes, and hence would be less than the time by which Z is exceeded by the time through BE . Let the point N be so chosen. It follows then that the time through BE after NB will be larger than the time Z , and thus will have a ratio to the indicated time through BI , with half the velocity from $B\Theta$, which is larger than the ratio of the arc FH to the line FG . Hence this ratio would be equal to the ratio of the arc FHO to the line FG .

Divide FG into the equal parts FP, PQ , etc., each of which is less than the height of the line NB and also less than the height of the arc HO . It is clear that this can be done. Next from the points of these divisions draw the straight lines $PA, Q\Xi$, etc., parallel to the base DC and ending at the tangent $B\Theta$. From the points where these lines cut the circumference FH , and also from H , draw the tangents $\Delta X, \Gamma\Sigma$, etc., upwards as far as the next parallel. Likewise, from the points where the indicated parallels cut the cycloid, draw the tangents SV, TM , etc., upwards. After adding to the line FG one part GR equal to the parts of the division, draw $R\Phi$ parallel to DC . It is clear that this line cuts the circumference FHA between H and O because GR is less than the height of the point H above O . We will next argue as follows.

The time through the tangent VS with the uniform velocity acquired from BS is larger than the time of a continuously accelerated motion through the arc BS after NB . For the velocity from BS is less than the velocity from NB because the height of BS is less than the height of NB . But the velocity from BS is assumed to be uniformly continued through the tangent VS since the velocity acquired from NB is continuously accelerated through the arc BS , and this arc is less than the tangent VS , and all of its parts are more vertical than any part of the tangent VS . Therefore the time through the tangent VS with the velocity from BS will be clearly larger than the time through the arc BS after NB . Likewise the time through the tangent MT with the velocity from BT will be larger than the time through the arc ST after NS . And the time through the tangent IY with the velocity from BY will be larger than the time through the arc TY after NT . And thus the sum of the times of the uniform motions through all the tangents down to the lowest one which touches the cycloid at E , with velocities through each one being equal to the velocity acquired by falling from B up to these points of contact, will be larger than the time through the arc BE after NB . But these same times will also be smaller, as we will now show.

Consider again these same times of the uniform motions through the tangents of the cycloid. The time through the tangent VS with the velocity from BS is related to the time through the line BA , with half the velocity from FA , as the tangent to the circumference ΔX is related to the part FP of the axis [Proposition 23]. Likewise the time through the tangent MT with